A Quantum-Safe Orchestration Layer with Client-Side Encrypted Inference

Architecture, measured results, and a verification methodology that produced 21 defect classes

Technical Report, revision `a1cd874-dirty`

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Abstract

This report describes a research proof-of-concept that chains post-quantum key establishment, quantum key distribution simulation, secure multiparty computation and CKKS homomorphic encryption into a single orchestration layer, and evaluates machine learning across 5 industry use cases, four of them with a privacy-preserving computation. The public demonstrator performs its homomorphic linear algebra and its garbled-circuit non-linearities *in the browser*, so its cryptographic timings are measured client-side rather than asserted. 11 models are published with pinned revisions.

The contribution we consider most transferable is not the architecture but the verification methodology. Over several rounds it produced 21 distinct classes of silent defect — a constant-class predictor whose accuracy exactly equalled the majority-class rate, an inverted diagnostic label map that made the demonstrator display the opposite of the model's output, a shuffled-`DataLoader` label misalignment statistically indistinguishable from random, and a baseline cited from the correct paper but the wrong table, among others. Each was invisible to tests, type checks and aggregate metrics. Section 9 tabulates all 21 and states four in full, alongside four conclusions the authors published and subsequently retracted.

We report what the system measures, what it does not, and where its numbers are not comparable to the published baselines they sit beside.

Scope and status. Research proof-of-concept, not a production system. Real implementations where tractable, explicitly labelled simulation where not. Licence Apache-2.0. This document reports measurements produced by the code at revision `a1cd874-dirty`; every table is generated from the repository's own artefacts at build time rather than transcribed.

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1 Summary

This report documents a research proof-of-concept and, more usefully, the verification methodology applied to it. The system chains post-quantum key establishment, a quantum-key-distribution simulation, secure multiparty computation and CKKS homomorphic encryption, and runs 5 industry use cases entirely in the visitor's browser, four of them with a homomorphic or secret-shared step; the satellite page's anomaly score runs in plaintext. 11 models are published at pinned revisions.

Headline results: the encrypted tier matches its plaintext teacher to four decimal places on one use case (AUC-PR 0.9681 for both) and falls well short on another (AUC 0.7882 against 0.9147), so no single "encryption costs X" figure would be honest. 21 classes of silent defect were found in a system with green tests, and four previously published conclusions were retracted after further measurement. Structure follows ANSI/NISO Z39.18-2005 (R2010) — Summary, Introduction; Methods, Assumptions and Procedures; Results and Discussion; Conclusions — with one deliberate departure: the defects and retractions are given their own section rather than being folded into Limitations. No standard blesses a "negative results" heading, and Limitations is the conventional home for them; they are separated here because they are a principal result of this work rather than a qualification on one.

2 Introduction and scope

2.1 Problem

Post-quantum migration is a procurement deadline before it is a research problem. NIST FIPS 203, 204 and 205 have been final since August 2024; NIST IR 8547, final since October 2026, deprecates every quantum-vulnerable public-key algorithm, RSA-2048, RSA-3072 and P-256 included, after 2030 and disallows them under NIST standards and guidelines after 2035, while the earlier deadlines of EO 14412 for federal high value assets and high impact systems stand; and for national security systems only, CNSSP 15 requires CNSA 2.0 algorithms in new acquisitions from 1 January 2027. Harvest-now-decrypt-later makes the confidentiality deadline earlier than the signature deadline for any data with a long secrecy lifetime.

Separately, privacy-preserving computation has become tractable for *small* models on a single device, and for large ones on GPU servers: Park et al. (arXiv:2601.18511) summarise 128 encrypted tokens of Llama-3-8B in 20 s on eight GPUs. Homomorphic evaluation remains cost-bounded: a recent systematisation puts private LLM inference at roughly four orders of magnitude slower than plaintext, and reports that only about 20% of surveyed systems are model-preserving. So a project that wants both a genuine industry model and genuine encryption in one browser tab has two honest options: shrink the model until it fits the cipher, or state plainly that the cryptographic demonstration runs on something smaller than the headline model.

2.2 What this system is

This project takes the second option and makes the cost measurable. Each use case carries two model tiers: a T2 *teacher* — the real architecture, plaintext — and a T1 *student*, small enough to evaluate under CKKS, distilled from the teacher's own output. The gap between them is reported as a first-class metric rather than elided.

2.3 What is new here, and what is not

This project's own verification rule is that a novelty claim must survive an adversarial search for prior art. Applied to its headline claim, the search found prior art for each half and none for the whole, and reporting only the flattering part would be the kind of omission the rest of this document exists to avoid.

MPC in a browser is not new. webSPDZ (IACR ePrint 2025/487) compiles MP-SPDZ to WebAssembly and communicates over WebRTC or WebSockets, and it names JIFF and MPyC-Web as earlier browser MPC engines. The garbled-circuit half of this system re-implements a known idea.

Homomorphic *evaluation* in a browser is not new either. node-seal, the WebAssembly build of Microsoft SEAL that this system uses, evaluates in the browser, and its own benchmark times evaluation operations in Chrome, Firefox and Safari; Attrapadung et al. (AsiaCCS 2018) shipped two-level homomorphic encryption compiled to WebAssembly. What the search did not find is narrower: a peer-reviewed 2025-26 report of encrypted neural-network inference evaluated in a browser tab. The field's default division of labour is the opposite one. IACR ePrint 2026/515 defines "client-side operations" as encryption and decryption while *users offload large computations to servers*; Zama's TFHE-rs WebAssembly API documents support for key generation, encryption and decryption but *not* FHE computation, citing WASM threading and a 2 GB memory ceiling. This system evaluates a real CKKS dot product and a real CKKS convolution in the tab.

The claim is therefore the composition rather than either half: PQC key establishment, then a simulated QKD link (it yields a key length, not key bytes), then secure multiparty computation, then homomorphic evaluation, then model inference — chained end to end in one browser tab with no server-side compute. A search for an integrated browser system performing that whole chain returned nothing. Two caveats travel with the claim and are stated wherever it appears: the two MPC parties are notional, running over an in-memory transport within a single tab, so the protocol, the bytes and the wall-clock are real while the party separation is not; and the quantum channel is simulated, because no browser API for quantum communication exists. Client-side execution is also not automatically private — WWW 2026's WAMO paper argues that browser deployment moves a model into a hostile environment, which is among the reasons the models here are published openly rather than treated as secrets.

2.4 What this system is not

It does not claim quantum *advantage*. The measured quantum-kernel and variational classifiers lose to a RandomForest baseline on the same data, and the registry records `quantum_beats_classical: false` rather than omitting the comparison. The trained QCNNs are trees of two-qubit blocks on 7 to 9 qubits, which a classical computer simulates exactly; they sit in the regime Bermejo et al. show to be classically simulable for classical data in a product angle embedding (PRX Quantum 7, 020304, 2026, Sec. IV.B). Nothing on their path is encrypted, and logistic regression beats each of them on the same split. The value proposition is quantum-*safety*, which is regulation-driven and dated.

3 System architecture

3.1 Layers

Four workstreams share one pipeline and a crypto-agility registry:

- **WS1 — AI-QKD.** BB84 two-intensity decoy-state simulation with GLLP tagged-fraction accounting, a selectable E91/CHSH protocol, a simulated quantum-measurement QRNG (Born-rule draws from a pseudo-random generator by default, with an opt-in online hardware source in Python) with SP 800-90B continuous health tests and a Leftover-Hash-Lemma extractor.
- **WS2 — QML intrusion detection.** Eight classes (seven QKD attack types plus normal traffic), scored on macro F1 rather than attack-versus-normal F1.
- **WS3 — QSMPC.** Shamir secret sharing with d-level QFT secure summation, a simulated

quantum oblivious transfer, and a keyed-hash set intersection over the PQC-protected channel, which does not hide the two parties' lists from each other (see the finance use case).

- **WS4 — QHE and AI.** CKKS linear layers with MPC non-linearities, including a genuine homomorphic convolution (im2col into `dot_plain`); in the browser the convolutional model's ReLU and pooling run in plaintext between its encrypted layers.

3.2 Crypto-agility as an architectural property

Every algorithm is resolved from a registry and selected before a run. There is no in-band negotiation and no silent fallback: this is crypto-agility in the EU NIS Cooperation Group sense — modular design enabling component replacement — and deliberately not cipher-suite negotiation during protocol execution.

Table 1: Cryptographic parameters as configured

Layer	Primitive	Note
KEM	`ML-KEM-768 + X25519`	FIPS 203 hybrid; ct 1088 B
Signature	`ML-DSA-65`	FIPS 204; sig 3309 B
FHE	`CKKS, ring degree 8192`	4096 slots, one multiplicative level per dot product
MPC	`Garbled circuits (Polytune)`	non-linear layers: ReLU, argmax
QKD	`BB84 two-intensity decoy`	GLLP tagged fraction

3.3 The slot budget, which is the real constraint

A CKKS ring of degree 8192 provides 4096 slots, and a ciphertext-plaintext dot product costs one multiplicative level. Parameter count is therefore *not* the binding budget; slot width is. Section 4.4 shows what that distinction bought once it was recognised.

4 Machine learning methodology

4.1 Teacher-student distillation, and what is reported

Students are fitted against the teacher's own output rather than ground truth, so `agreement` measures substitution rather than two independent fits that happen to correlate. Fidelity and task performance are reported separately because they decouple empirically: in one case agreement rose from 0.8841 to 0.9471 in Round 24 while the task metric did not move at all. The current value is 0.9829; the pair is quoted as a before/after, not as today's figure. Reporting one as though it were the other is a category error this project made and corrected.

4.2 Metric selection

Four of the five verticals have a documented failure mode in which the obvious aggregate looks healthy while the class that matters is not learned. Each headline metric is therefore the one the relevant literature reports, with the flattering companion printed beside it:

- **Finance:** minority-class F1 at a 1-in-981 positive rate. The precision-recall baseline is $P/(P+N) \approx 0.001$, so an ROC-AUC of 0.95 says nothing.
- **Satellite:** event-wise $F_{0.5}$ under an overlap rule. Point-adjusted F1 is computed *only* to show its inflation — under point adjustment "even a random anomaly score can easily turn into a state-of-the-art TAD method" (Kim *et al.*, AAAI 2022).
- **Medical:** AUC, with the majority-class rate and the number of distinct classes predicted printed alongside (Section 9 explains why).

4.3 Operating-point selection by net benefit

The screening operating point is chosen by net benefit,

$$\mathrm{NB}(p) = \frac{\mathrm{TP}}{N} - \frac{\mathrm{FP}}{N} \cdot \frac{p}{1-p},$$

in which the subtracted term is the exchange rate the threshold itself declares. Three criteria were tried and two rejected on measurement:

- **F1 — rejected, degenerate.** When discrimination is weak, minority-class F1 is maximised by calling everything positive: recall goes to 1 while precision falls only to the base rate. The criterion selected the extreme grid point and the resulting model predicted the positive class for all 156 test images. The validation score it chose, 0.4242, is exactly the all-positive value $2p/(1+p)$ at $p = 0.269$. This is a theorem, not a local surprise — Lipton et al. (2014) prove that an uninformative classifier's F1-optimal behaviour is to classify every example as positive.
- **Youden's J — rejected, symmetric.** J cannot be gamed the same way, but it is the symmetric special case: it prices a missed cancer and a false alarm equally, which no screening programme does and which this project had never stated.
- **Net benefit — adopted.** It makes the exchange rate explicit and places the model on the same axis as treat-all and treat-none, so "worse than not screening" is visible rather than inferred.

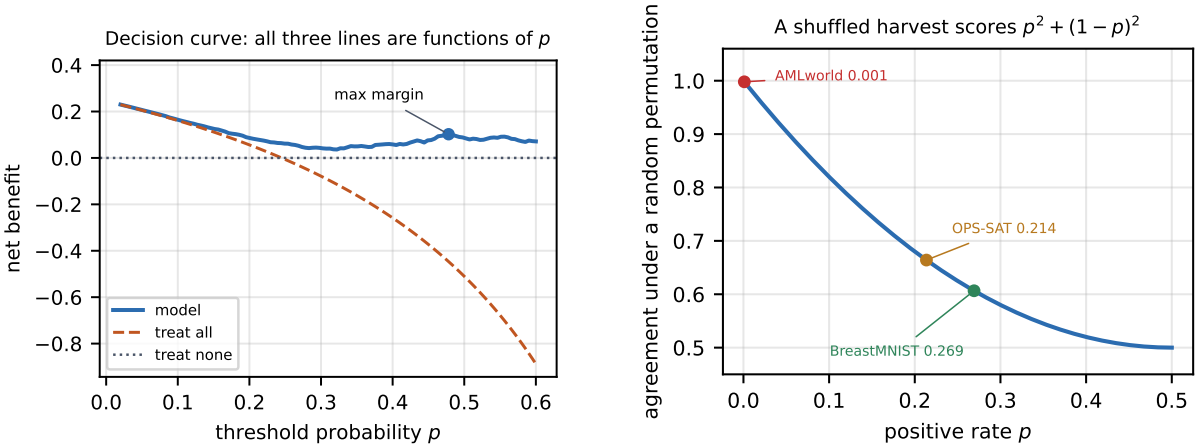


Figure 1: Left: a decision curve. All three lines are functions of p, and only matched-p comparisons are meaningful — an error the authors made and corrected (Section 9, W8). **Right:** agreement expected from a randomly permuted label harvest, $p^2 + (1-p)^2$. At a 1-in-981 positive rate a completely broken pairing scores 0.998, which is why an alignment check with a "near 1.0" pass condition cannot detect misalignment on an imbalanced task.

4.4 Client-side feature expansion under encryption

A polynomial expansion of the input is a function of the *plaintext*, so the client computes it before encrypting and the encrypted step remains a single packed dot product at unchanged multiplicative depth. For the OPS-SAT segment classifier this expands 19 features to 209 — well inside the 4096-slot budget — and measurably improved the encrypted tier:

	linear	quadratic	paired bootstrap 95% CI on the delta
minority-class F1	0.8649	**0.9043**	[+0.0154, +0.0929]
AUC-PR	0.9275	**0.9681**	[+0.0089, +0.0853]

Both intervals exclude zero over 4000 paired resamples, so the gain survives at this sample size. The interval is wide — its lower bound is a fifth of the point estimate — so the defensible claim is "a real improvement of at least about one AUC-PR point", not the point estimate.

Three qualifications belong with that result. The technique is *not novel*: the mathematics is a known explicit feature map, and placing feature construction at the client-side plaintext boundary specifically to bound homomorphic depth is published system design. Depth is unchanged but cost is not zero — slot-summation rotations rise from $\lceil \log_2 19 \rceil = 5$ to $\lceil \log_2 209 \rceil = 8$ key-switches. And the client must know the feature map, which discloses model *structure* though not weights.

Applied to the finance student the same change *lowered* the headline metric, so it is a per-model flag rather than a default. That negative result is recorded in the flag's own help text.

5 The five use cases

Each use case exercises the full chain — post-quantum key establishment, key distribution, a privacy-preserving computation, and an inference — but they were chosen so that the *computation* differs, not merely the data. A demonstrator in which five verticals run the same pipeline over five CSV files would show nothing about the cryptography.

5.1 Finance: anti-money-laundering with cross-institution set intersection

Two institutions need to know whether a counterparty appears on both watchlists without disclosing either list. The demonstrator's set intersection is an HMAC of both lists under the one session key both banks hold, so over a small identifier space either bank can hash every candidate identifier and recover the other's list: the naive-hashing failure Pinkas, Schneider and Zohner describe (USENIX Security 2014). What holds is that no third party sees either list and that the channel is protected by the PQC session; a production deployment would use an OPRF-based PSI. The encrypted inference in the browser is a logistic fraud score under CKKS, over eight features of the bundled PaySim sample. The published teacher and student are trained on IBM AMLworld HI-Small: 5,078,345 transactions at one laundering event in 981, evaluated on a strictly temporal split with the decision threshold tuned on a third window the model was never fitted on.

The gap to the published baseline is the largest in the project and its cause is architectural rather than a tuning failure. The reference implementation computes per-transaction subgraph patterns over a dynamically updated graph; this project computes per-account scalar aggregates. A degree is one number per account; the reference feature is a pattern match in that transaction's own neighbourhood at the moment it arrives. An earlier note attributed the gap to a proprietary licence and to truncated multi-hop reachability. The licence is real but was never the mechanism, and the reachability claim did not survive the source's own ablation: its Table 5 (HI-Small, XGBoost, batch size 128) gains 37 F1 points from fan-in and fan-out patterns over basic features, 19.75 to 56.88, and 1.7 more from multi-hop cycle and scatter-gather patterns, to 58.60. This project already computes fan-in and fan-out, as per-account aggregates.

Investigating that gap found a genuine defect. Per-account aggregates were computed over the whole training window, so a training row's feature included activity that happened after it while test rows received the same table as a stale snapshot. Making the aggregates causal — each row sees only strictly earlier edges — raised minority-class F1 from 0.0808 to 0.1519. Leakage was disproved rather than argued: every feature is bit-identical whether or not later rows exist.

5.2 Healthcare: federated training with encrypted diagnosis

Hospitals train jointly without pooling records, so the primitive is masked aggregation of federated updates: pairwise masks that cancel exactly in the sum. In this demonstrator every mask is derived from the one PQC session key, which the aggregator also holds, so the masks hide nothing

from it; Bonawitz et al.'s secure aggregation derives each pair's mask from a key agreement between that pair alone (arXiv:1611.04482). The inference includes a genuine homomorphic convolution, im2col into CKKS dot products. In the browser each layer is decrypted in the client between steps: ReLU and pooling run there in plaintext, and the convolutional model's argmax is taken in the clear, while the garbled-circuit argmax runs on the Wisconsin logistic regression. This is the only use case where the encrypted path evaluates a convolutional layer rather than a linear model alone.

It is also where the most serious defects were found, and all of them were in the *measurement* rather than the cryptography: a constant-class predictor, an inverted diagnostic label map, and a label/sampler misalignment. Section 9 treats them in full. The current encrypted student reaches AUC 0.7882 against its teacher's 0.9147, with accuracy above the majority-class rate and both classes predicted.

5.3 Genomics: pooled allele counts by secure summation

Research sites contribute allele counts without revealing per-site genotypes: additive secret shares in the browser, d-level QFT secure summation in the full deployment. The encrypted step is a sign test on 8 of the 313 PGS000004 variants, not the whole polygenic score, which the tab computes in plaintext. This use case deliberately reports *no* AUROC: the reference panel carries no case/control status, so the quantity is undefined, and printing a plausible number for it would be the single most tempting error available here. What is computable from real genotypes is the ancestry portability gap, and that is reported instead, because it is the clinically important finding rather than a substitute for the missing one.

5.4 Telecom and defence: intrusion detection on quantum-channel telemetry

Eight classes — seven attack types plus normal traffic — scored on macro F1, because attack-versus-normal F1 would let a detector that recognises one common attack look competent. Both a classical RandomForest and quantum classifiers are trained on identical data, and the classical baseline wins: macro F1 0.9866 against 0.8316 for the quantum-kernel SVC and 0.4667 for the variational classifier. The registry records `quantum_beats_classical: false`. Reporting this is the point of running both. The published `telecom_ids.qml` artefact, the model the page runs and verifies, is that RandomForest, exported to ONNX; the two quantum classifiers are trained offline and not run by the page, and the SVC's fidelity kernel equals the classical product kernel $\prod_i \cos^2((x_i - x'_i)/2)$.

5.5 Satellite: free-space QKD with direct telemetry

A simulated downlink under Micius's published loss; a QSDC check that computes the secrecy capacity of the modelled link with and without an eavesdropper; and anomaly detection in plaintext in the tab, a linear score over NASA SMAP and MSL telemetry windows beside the published forecaster trained on multi-year ESA telemetry. No homomorphic or MPC stage runs on this page. The secrecy criterion is the one place where a familiar constant proved actively misleading. A threshold of 0.11 reproduces the textbook loss-free case, and the deeper derivation gives a capacity

$$C_s \;=\; Q_{\mathrm{Bob}}\big[\,1 - h(e) - g\,h(e_x + e_z)\big], \qquad g = 10^{\,\mathrm{loss_dB}/10},$$

in which the loss term dominates; g counts the channel and the detector, not the station's optics. On the 800 km downlink (32.2 dB of channel loss, 27.3 dB from Eve to Bob's detector, $g \approx 533$) the admissible error rate falls from 0.11 to about 6.5×10^{-5} — three orders of magnitude, and in the permissive direction, so the constant would have declared links secure that are not. The scalar was replaced by the computed capacity.

The modelled downlink is Micius's, read value by value from Liao et al. (2017), with the atmosphere scaled by air mass from 5.5 dB at 1200 km and no link below 10 degrees of elevation.

Integrated over Liao's pass of 19 December 2016 and counted as the paper counts — signal states only — it gives the measured detections and sifted bits within 1 percent. A second tab models what carries a key past the horizon: the trusted satellite relay Micius flew between Xinglong and Graz, and a plane of satellites linked to their neighbours.

The key is the composable finite key of one pass peaking at the page's range, the block every flown experiment found computes its key on: Lim et al.'s (2014) bound for a source that sends the vacuum, on the pass's counts summed second by second, with each count's expectation bounded by Yin et al.'s (2020) Chernoff-type bound, which on a pass gives slightly more key than the union of it and the published Hoeffding deviation that the fibre uses. On Liao's pass at their failure probability of 10^{-9} it gives 226,072 bits, 0.75 of the 300,939 they report, and 164,183 bits, 0.55, with Hoeffding's deviation alone, as Lim et al. published it. On the Micius–Zvenigorod session, whose key was computed with Chernoff bounds, it gives 0.69 of the 629,000 bits Khmelev et al. report before their detector-mismatch analysis, and 0.56 with Hoeffding's: the statistics close 45 percent of the first gap and 29 percent of the second, and the two fractions part, so the rest is not one shared cause. With the Hoeffding statistics the papers used, the same bound reproduces Rusca et al.'s (2018) two-decoy figures to within 20 percent, always below them, and their ratio to the one-decoy figures within 15 percent; their one-decoy 26 dB row is reproduced to within 2 percent under the paper's own dead-time convention. Pooling passes, or a time held at one range, are offered beside it; at 800 km a key needs about 7 s of link time held at that range with the tighter of the two, and 33 s with Hoeffding's deviation alone. These are figures of the channel model, not of a link: no satellite, ground station or quantum channel is involved.

6 Implementation and verification architecture

This section shows how the system is built and checked before the results say what it measured. Figure 2 is the verification environment: what runs on a workstation, on GitHub and on the demo host, and the live check that closes every change. Figure 3 is the order a change goes through them. Figures 4 to 8 then follow each use case through its stages, naming what implements each stage. The README carries the same figures as Mermaid diagrams, and a test holds the labels of the two copies identical, figure by figure.

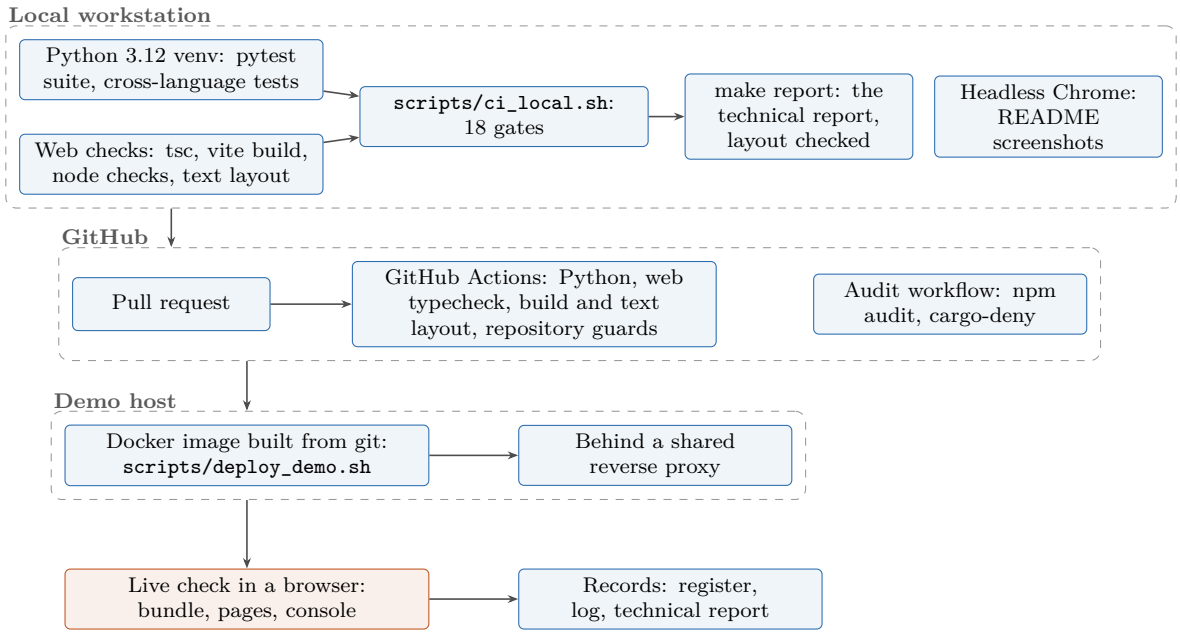


Figure 2: The verification environment. Every change runs the local gates, then GitHub Actions, then is deployed from git to the demo host and checked in a browser on the deployed site; the records close it. The demo host's address is deliberately not given.

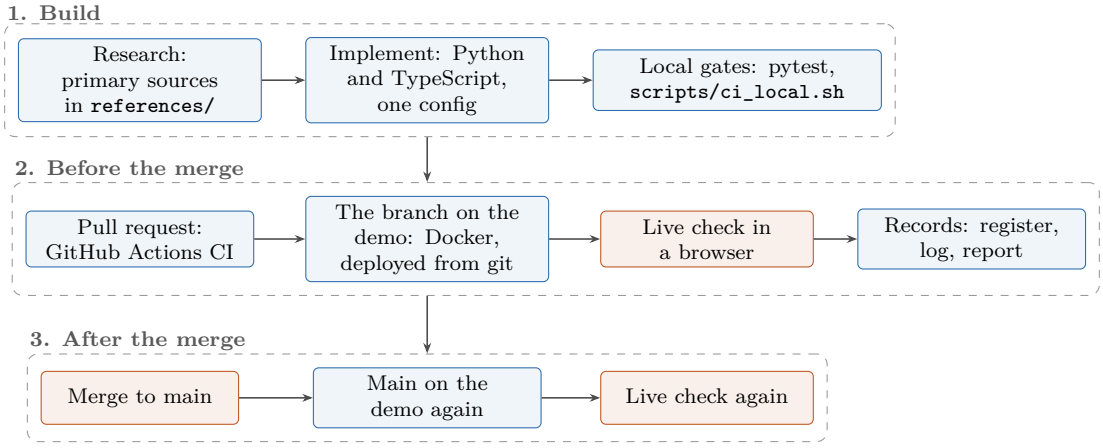


Figure 3: The implementation workflow. A branch is deployed and checked live before it merges, and main is deployed and checked again after; the merge itself is the owner's decision.

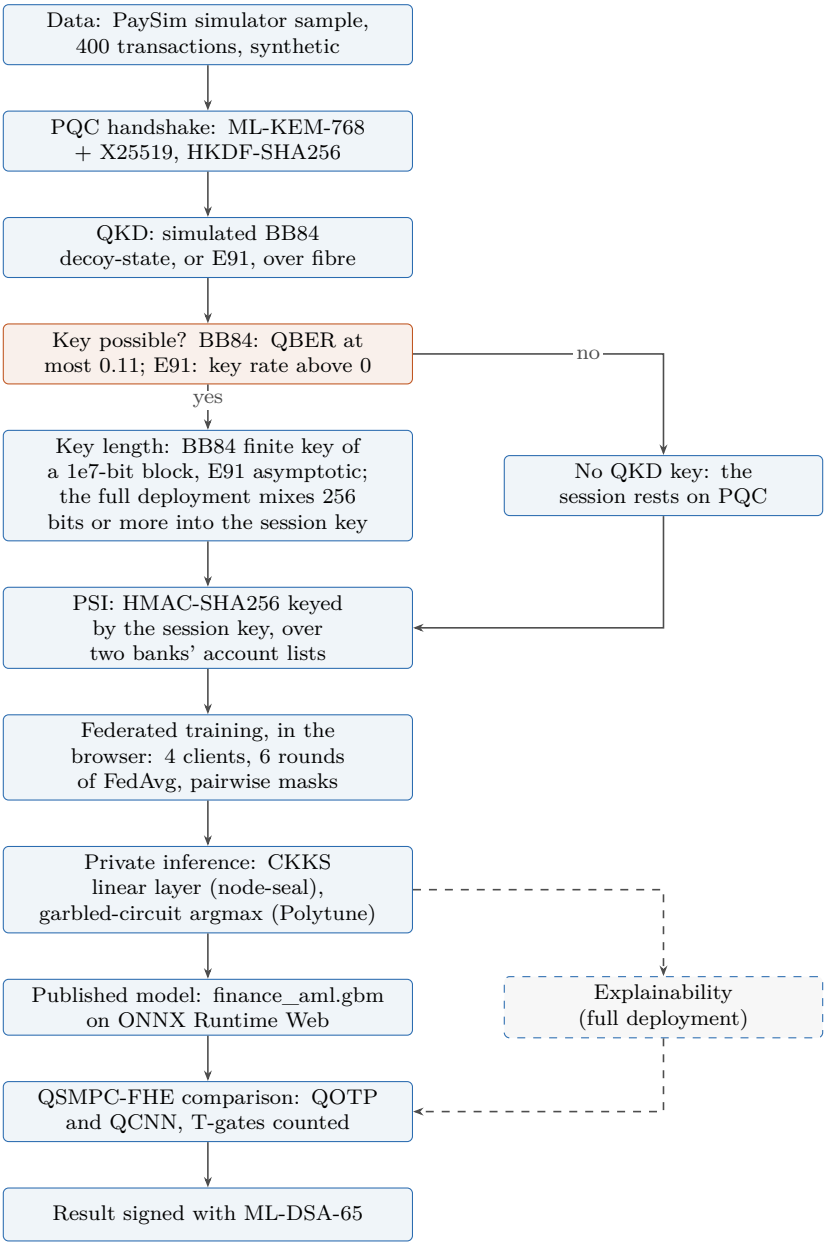


Figure 4: Finance: cross-bank anti-money laundering: the stages in the order the browser runs them, and what implements each. Dashed boxes run only in the full deployment.

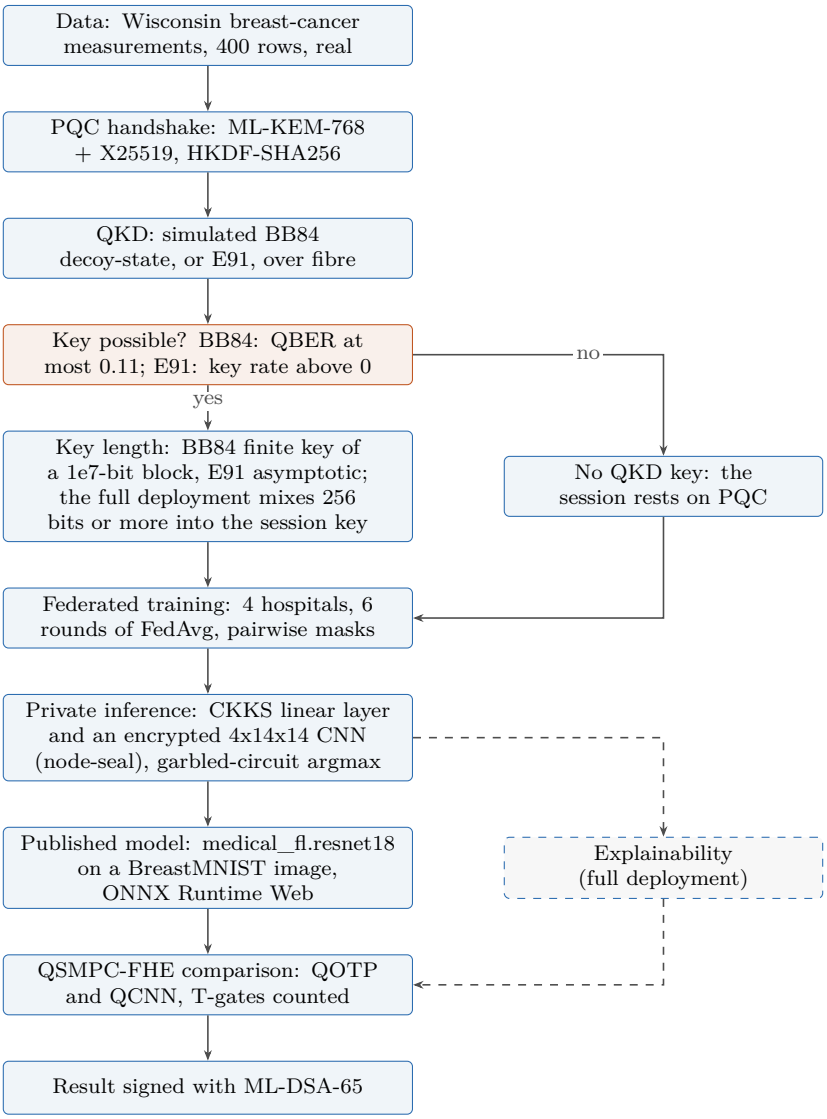


Figure 5: Healthcare: federated training and private diagnosis: the stages in the order the browser runs them, and what implements each. Dashed boxes run only in the full deployment.

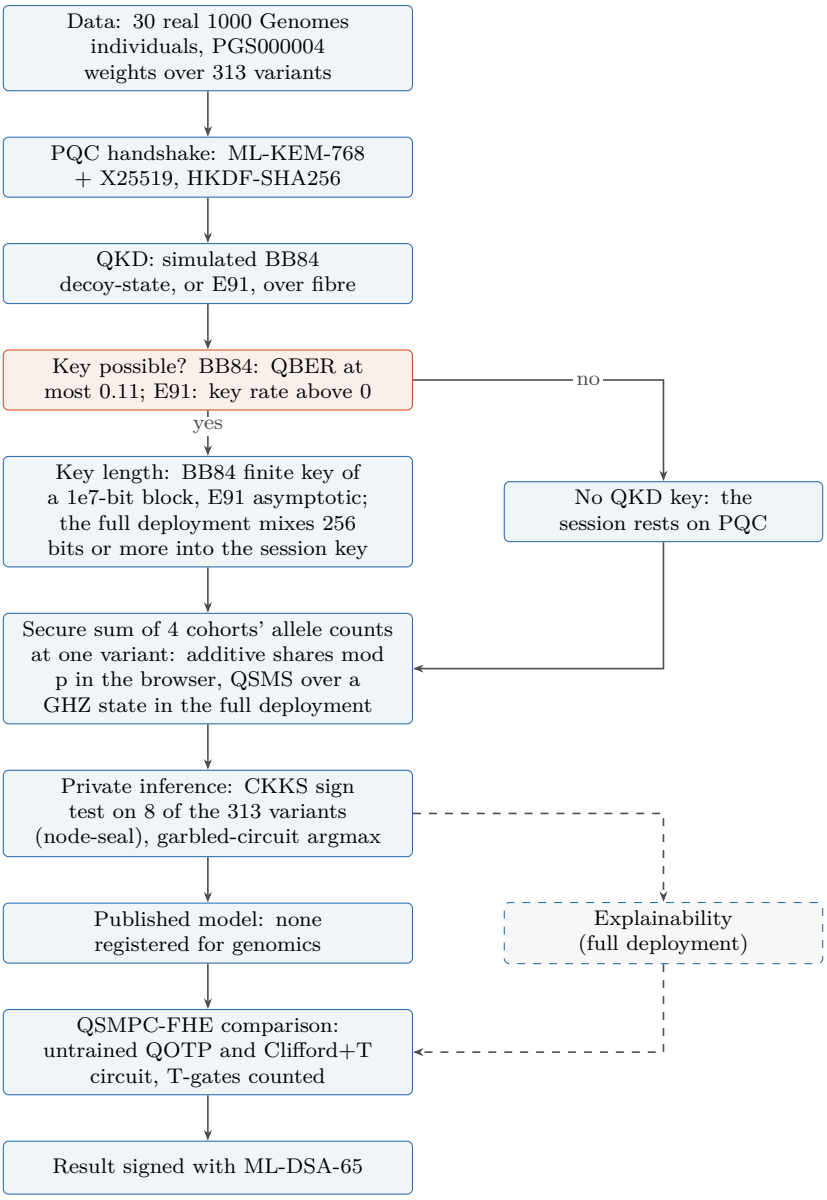


Figure 6: Genomics: pooled allele counts: the stages in the order the browser runs them, and what implements each. Dashed boxes run only in the full deployment.

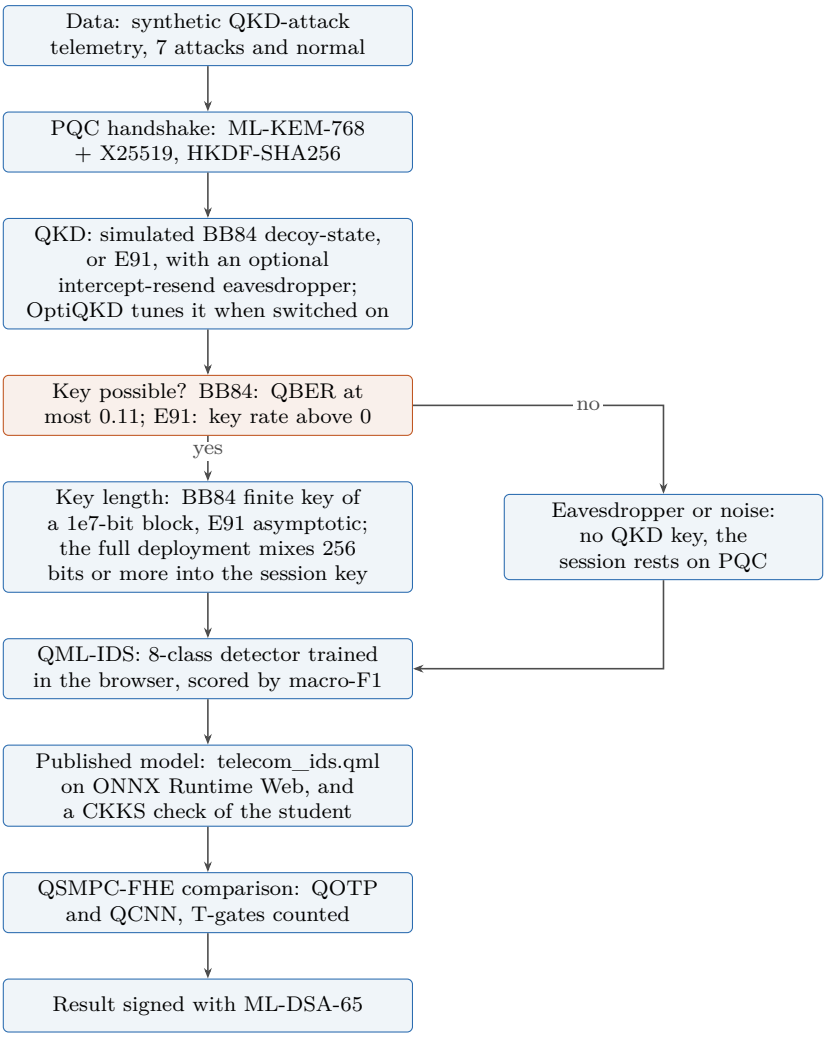


Figure 7: Telecom and defence: intrusion detection on a QKD link: the stages in the order the browser runs them, and what implements each.

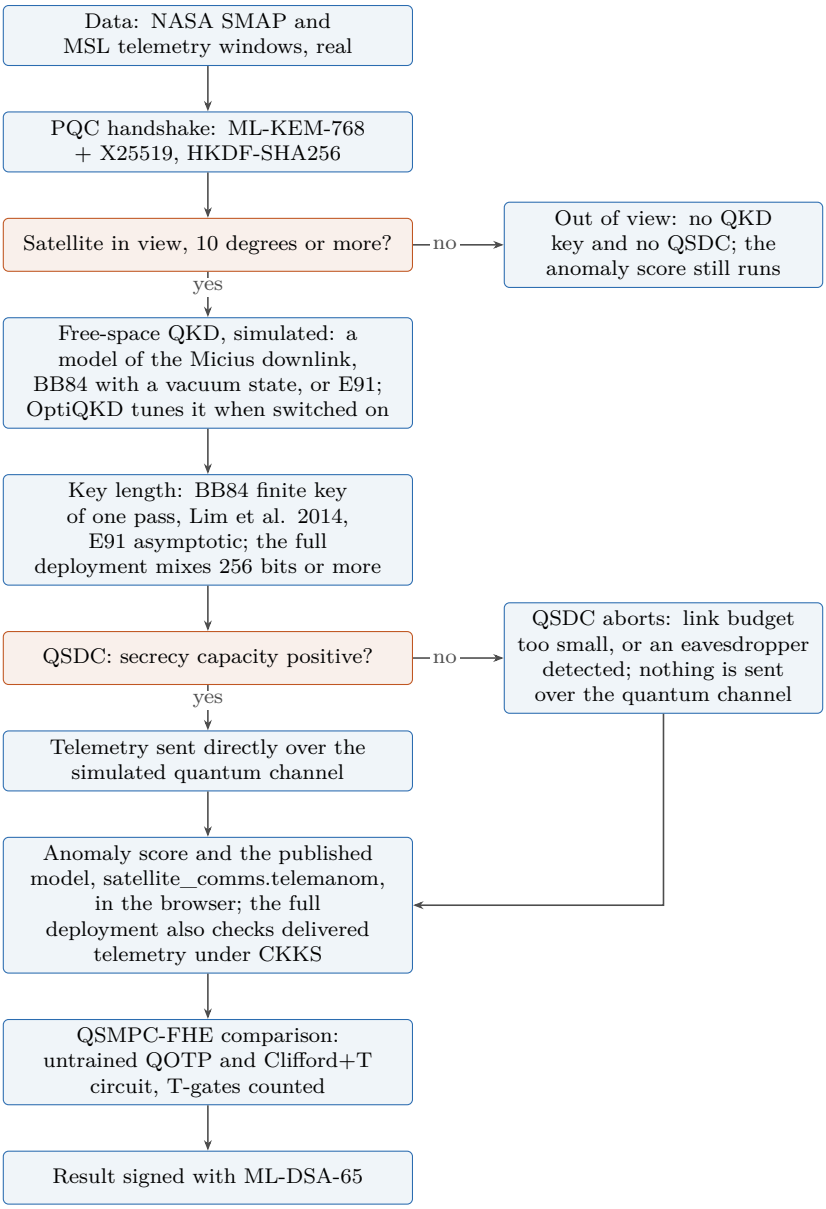


Figure 8: Satellite: free-space QKD and direct telemetry: the stages in the order the browser runs them, and what implements each.

7 Security model and its boundaries

7.1 What is real and what is simulated

The distinction is load-bearing and is stated per component rather than in aggregate:

Component	Status	Basis
ML-KEM-768 + X25519	real	library implementation, conformance-tested
ML-DSA-65 signatures	real	round-trip and tamper vectors
CKKS linear layers	real	node-seal, measured in the browser
Garbled-circuit argmax	real	Polytune 2PC, notional parties in one tab
BB84 decoy-state	*simulated*	channel model, not hardware
E91 / CHSH	*simulated*	channel model, not hardware
Satellite downlink	*simulated*	Micius's loss, Lim et al. (2014) finite key
QSDC secrecy capacity	*simulated*	closed form on the modelled loss
QRNG source	*simulated*	seeded pseudo-random draws, real health tests
QOTP and QCNN circuits	*simulated*	state vector, classically simulable

Nothing labelled simulated is reported with a wall-clock timing that implies otherwise. The public demonstrator performs no server-side computation at all, which is why its homomorphic and MPC timings are meaningful: they are measured in the visitor's own tab.

7.2 Conformance rather than happy-path testing

Post-quantum library correctness is checked with negative vectors, not only round-trips. This distinction was not academic. An earlier harness reported ten of ten passes on a negative-vector suite whose negatives were wrong-*length* inputs — a test that any implementation passes by rejecting malformed input, and which says nothing about arithmetic. Modulus vectors were decisive where the original suite was not. Separately, a library upgrade reversed an argument order in its signing API; type checking, the build and the full test suite all passed, and only the browser surfaced it.

7.3 Dependency currency, and a gap no scanner reports

The homomorphic layer uses `node-seal`, a WebAssembly build of Microsoft SEAL. The pinned dependency resolves to `node-seal` 7.0.0, whose vendored source declares **SEAL 4.1.2, released 11 July 2024**. Upstream shipped **4.4.0 on 25 July 2026** described by its maintainers as a critical security update hardening untrusted deserialization, then 4.4.1 and 4.4.3 in the first week of August 2026. *No CVE or advisory exists for either package*, so dependency scanners report nothing and the lockfile is current against its own registry. The gap is findable only by reading upstream release notes.

Whether it matters here depends on the architecture, and the answer is specific rather than reassuring-by-default. The hardened surface is deserialization of untrusted input. This system generates keys, encrypts, evaluates and decrypts entirely in the visitor's own tab, performs no server-side computation, and deserialises no externally supplied ciphertext — so that surface is not reachable in this deployment. It would become reachable the moment any server component accepted a ciphertext from a client, and that is recorded as a condition rather than a footnote.

The same reasoning governs a second upstream statement. SEAL's security documentation was revised in July 2026 to state that its CKKS implementation targets IND-CPA and **not** IND-CPA-D, and that noise flooding must be applied by the application before a decrypted CKKS value crosses a trust boundary. Here the secret key, the decryption and the plaintext result are all in the same tab, so no such boundary is crossed. The architecture is what makes this safe; the cipher alone does not.

7.4 Known limitations of the threat model

The client-side feature map used in Section 4.4 discloses model *structure* — the identity of the expansion — though not weights. Published work treats exactly this as a problem worth solving with randomised shuffling. Where model confidentiality is part of a claim, the disclosure must be stated; here it is, and the claim is not made.

8 Results

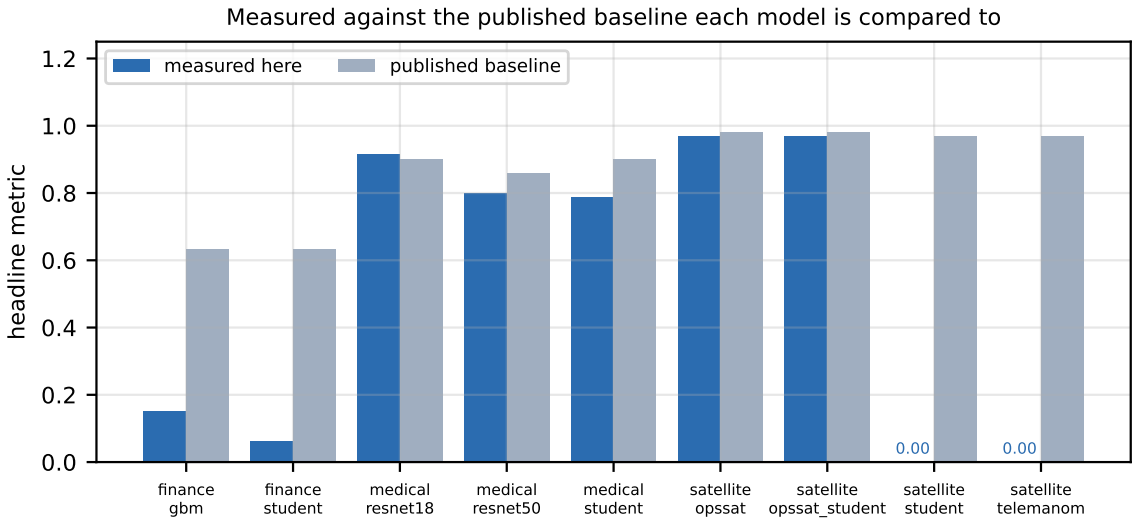


Figure 9: Measured value against published baseline. Two comparisons are deliberately not like-for-like and are discussed below rather than presented as parity.

8.1 Where the numbers are not comparable, stated explicitly

- **Satellite event-wise $F_{0.5}$ is computed over three labelled events.** The corpus contains 320 annotated intervals over the full timeline, but the annotated anomalies are concentrated early and a temporal split leaves three inside the scored window. The current forecaster, retrained in Round 71 with its normalisation fitted on the training window alone, scores 0: it flags 2 points and neither falls inside a scored event (tp 0, fp 2, fn 3). The two runs before it, trained with a whole-series normalisation that leaked the test window, scored 0.714 and 0.455, which decompose uniquely on three events and differ by exactly one extra false-alarm run. The reference partition holds 65 events. The number is reported with its denominator attached.
- **The reference metric is corrected; ours was not.** The benchmark multiplies event precision by the time-level true-negative rate, which is why most of its baselines read below 0.001. Both values are now computed.
- **Genomics reports no AUROC.** The reference panel has no case/control status, so the quantity is undefined and is not printed. What *is* computable from real genotypes is the ancestry portability gap, which is the clinically important finding.

8.2 The encrypted tier's cost

The distilled students are 33 to 3,226 parameters. For the medical use case the encrypted student reaches AUC 0.7882 against its teacher's 0.9147; for OPS-SAT the encrypted student matches its

Table 2: Published models: measured value against the baseline each is compared to. Tier T1 runs under encryption; T2 is the plaintext teacher. A dash means the model is declared in the registry but not trained in this build, which is reported rather than hidden. Dataset licences are given in full in the next table; this column names the corpus so the licence is never abbreviated.

Model	Tier	Metric	Measured	Baseline	Dataset
`finance_aml.student`	T1	minority-class F1 (laundering)	0.0614	0.6323	IBM AMLworld HI-Small
`finance_aml.gnn`	T2	minority-class F1 (laundering)	–	0.6323	IBM AMLworld HI-Small
`finance_aml.gbm`	T2	minority-class F1 (laundering)	0.1519	0.6323	IBM AMLworld HI-Small
`genomics.prs`	T2	AUROC (European ancestry)	–	0.6330	PGS Catalog scoring files
`medical_fl.student`	T1	AUC	0.7882	0.9010	MedMNIST v2 BreastMNIST
`medical_fl.resnet18`	T2	AUC	0.9147	0.9010	MedMNIST v2 BreastMNIST
`medical_fl.resnet50`	T2	AUC	0.7988	0.8570	MedMNIST v2 BreastMNIST
`satellite_comms.student`	T1	event-wise F0.5	0.0000	0.9680	ESA Anomaly Dataset (ESA-ADB), Mission1 lightweight channels 41-46
`satellite_comms.opssat_student`	T1	AUC PR	0.9681	0.9790	OPS-SAT-AD
`satellite_comms.telemanom`	T2	event-wise F0.5	0.0000	0.9680	ESA Anomaly Dataset (ESA-ADB), Mission1 lightweight channels 41-46
`satellite_comms.opssat`	T2	AUC PR	0.9681	0.9790	OPS-SAT-AD
`telecom_ids.student`	T1	macro F1 (synthetic QKD-attack telemetry)	0.9883	–	QKD attack telemetry (synthetic, methodology-grounded)
`telecom_ids.transformer`	T2	macro F1 across attack classes	–	–	CIC-IDS2017, Engelen-corrected redistribution (WTMC 2021)
`telecom_ids.qml`	T2	macro F1 (synthetic QKD-attack telemetry)	–	–	QKD attack telemetry (synthetic, methodology-grounded)
`telecom_ids.nidd`	T2	macro F1 across 9 network-attack classes	0.9950	–	5G-NIDD (BTS1_BTS2_fields_preserved.csv)

Table 3: Dataset licences, quoted whole. An earlier revision printed this column inside the models table through split(";")[0][:22], which removed the qualifier from three of the seven licences that reach this report - and a qualifier is what makes a licence string long, so truncation is biased toward overclaiming.

Dataset	Licence, in full
5G-NIDD (BTS1_BTS2_fields_preserved.csv)	CC-BY-4.0
CIC-IDS2017, Engelen-corrected redistribution (WTMC 2021)	provider's own pages state conflicting terms
ESA Anomaly Dataset (ESA-ADB), Mission1 lightweight channels 41-46	CC-BY-3.0-IGO
IBM AMLworld HI-Small	CDLA-Sharing-1.0
MedMNIST v2	CC-BY-4.0 asserted by MedMNIST; the BUSI authors publish no licence
OPS-SAT-AD	CC-BY-4.0
PGS Catalog scoring files	per-score; EMBL-EBI Terms of Use by default
QKD attack telemetry (synthetic, methodology-grounded)	no licence granted; synthetic rows generated by this repository's own code

teacher's AUC-PR to four decimal places, 0.9681 (0.968055 against 0.968107). Both figures score the student in plaintext, offline; how often CKKS noise flips a decision is measured separately, in the browser, on every run. The gap is therefore strongly task-dependent, and reporting a single "encryption costs X percentage points" figure across use cases would be misleading.

9 What we got wrong, and how it was found

This section exists because the defects below were invisible to every automated signal the project had. Shapes matched, losses decreased, tests passed, type checks passed, and aggregate metrics looked plausible. Each was found by a specific act of measurement, and the check that would have caught it is now a numbered row in the verification checklist (673 rows at this revision).

Table 4: 21 classes of silent defect, all found in a working system with green tests

Class	What it was	Why nothing caught it
Constant predictor	medical CKKS student emitted one class for all 156 test images	accuracy 0.7308 equalled the majority rate exactly; AUC 0.354, below random
Inverted class labels	four sites asserted the opposite of MedMNIST's own metadata	the demo displayed the opposite diagnosis to the model's output
Sampler misalignment	shuffled DataLoader output paired positionally with dataset order	teacher/ground-truth agreement 0.6227, inside the permutation null
Wrong baseline row	cited ESA-ADB Table 2 (all events) for an anomalies-only task	like-for-like row is Supplementary Table 9 at 0.968, not 0.786
Uncorrected metric	reported uncorrected event-wise F0.5 against a corrected baseline	ESA-ADB multiplies event precision by the time-level TNR
Licence divergence	published artefact recorded ImageNet weights; two docs said random	publish now refuses any model whose manifest records a pretrained source
Guard too narrow	both_scored_zero required BOTH models at zero	teacher 0 vs student 0.0032 rendered as the encrypted model beating its teacher
Manufactured margin	treat-all evaluated at a different threshold from the model	a zero margin reported as +0.2308; caught by the degeneracy guard
Leak in a published sample	finance eval sample: hour_of_day >= 3 reproduced the label at 100% precision	served from an HF revision whose pin looked valid; the first two leak guards were themselves wrong
Guard exempted by its data	telecom feature_names held CLASS names, not feature names	the leak guard skipped the file and reported success on something it never examined

Class	What it was	Why nothing caught it
Half-applied fix	QCNN repaired in the browser; the Python path stayed a constant predictor	a coin flip looks broken, a constant predictor looks confident and reproducible
Crossed wiring	the trainer's CNOT ring indexed a stale axis after chained movedim	training still converged; only an independent reimplementation disagreed, by 0.73 on a quantity bounded in [-1,1]
Feature-space mismatch	circuit trained on min-max raw units, served StandardScaler z-scores	seven of eight features pinned to -1; a degenerate state still returned a confident class
Selection on the test set	epoch chosen and headline reported on the same rows	medical F1 fell 0.9181 to 0.7326 once a validation split was introduced
Criterion half-applied	loss-aware QSDC bound added to the module, not to its caller	an inert threshold control, and an eavesdropper reported on every clean run
False provenance	a panel announced that the quantum circuit produced a class CKKS had produced	it read the CONTROL value; the artifact recording which backend ran was never consulted
Tautological verdict	circuit_private rested half on decrypt(encrypt(x)) == x, which is 1.0 by construction	a passing test asserted both the tautology and the verdict, so green read as confirmation
Guard that cannot fail	r.ok tested on an SPA host that answers 200 with HTML for any path	it degraded correctly only because JSON parsing then threw and a catch swallowed it
Label true of another artefact	Clifford+T named a Python module containing zero T gates	the same label IS accurate for the browser circuit, which counts 56 of them
Held-out by assertion	three published evaluation samples were drawn from the whole corpus, training rows included	the log called every sample held-out; redrawn from the test window the finance teacher's rare-class F1 fell 0.5439 to 0.2032
Fix that missed the artefact	a z-score leak fixed in the trainer, and the published weights never retrained	their constants still matched the whole series; retrained, event-wise F0.5 fell to 0 on 3 events

9.1 Four findings worth stating in full

W1 — A model that had learned nothing, with a respectable number. The medical CKKS student emitted a single class for all 156 test images. Its accuracy of 0.730769 is exactly 114/156, the majority-class rate, and its AUC was 0.354 — below random. It was invisible because the registry declared AUC as the model's headline while the record stored accuracy: AUC would have been unmissable, accuracy read as "behind but plausible". The metric-name mismatch had been on the backlog as a cosmetic registry inconsistency. It was the thing hiding the failure.

W3 — Labels attached to the wrong images. Teacher labels were harvested through a `DataLoader` constructed with `shuffle=True` — correct for training, and the loader was also the training loader — then paired positionally with the dataset in its original order. Every distilled label landed on a different image. Measured teacher/ground-truth agreement was 0.6227 where a sequential pass gives 1.0000; 4000 simulated permutations give a 99% interval of $[0.5641, 0.6484]$, so the harvest was statistically indistinguishable from random. `predict_loader` now refuses a random sampler outright.

Two rounds had attributed this student's weakness to CKKS capacity — "3,226 parameters is a feasibility budget, not a diagnostic one" — and that explanation had been written into three documents. Training the identical architecture on aligned labels scored 0.23 AUC higher, which disproved it. The resulting rule: before concluding that the encrypted tier is small and therefore weak, train the same small model properly once and see what it can do.

W8 — A comparison that manufactured its own margin. The first implementation of net benefit scored the model across a threshold grid but evaluated treat-all *once* at the grid midpoint, reporting "best +0.2308 at $q = 0.050$ (treat-all −0.0150)". At $q = 0.050$ the model flagged every validation case, so it *was* treat-all, and treat-all at that same threshold also scores +0.2308. The true margin was zero and the comparison produced a win by moving the goalposts to a different p. It was caught because an unrelated guard — added earlier to reject any threshold collapsing to a single class — refused to fold the result.

W9 — A label leak inside a published artefact, behind a pin that looked valid. The evaluation sample published alongside the finance model contained a single feature that reproduced its label: every row with `hour_of_day` ≥ 3 was laundering, at 100% precision. Anyone who downloaded the sample to reproduce the reported figure would have scored a trivial threshold rule and read it as the model working. Three things had to fail together. The sample was regenerated after the pin was recorded, and nothing invalidated the pin, so `hf_revision` still resolved and still looked current. The guard written to catch exactly this was wrong twice — first testing disjointness of unique values, which is meaningless on continuous floats, then computing a binary rank AUC over multi-class labels and printing 5.1292, a number outside $[0, 1]$ that was not questioned. And the neighbouring use case escaped the guard altogether because its `feature_names` field held class names, so the guard iterated over nothing and reported success.

The repository's own headline figure was never affected: it is computed over 1,523,504 rows that the leaked sample never touched. An earlier draft of this section said the leak accounted for 13.8% of the model's headline number; that was wrong, and the 13.8% belonged to the verification figure. The correction matters more than the original claim, because a defect report that overstates its blast radius is the same category of error as the defect. What the episode establishes is narrower and worse: a guard that reports success without examining anything is indistinguishable, in a log, from a guard that passed.

9.2 Retracted conclusions

Four published conclusions were withdrawn after further measurement:

(a) **"Seed variance."** Two teacher AUCs were reported as moving on reruns "with no functional change", supported by a six-seed spread. The initialisation had changed, not the seed: the trainer defaulted to pretrained weights and the reruns omitted the flag. With it, both reproduce the documented figures to four decimal places.

(b) **"The student's ceiling is 0.0262."** A threshold sweep was reported as proving capacity was the binding constraint, and a probe predicting three times that performance was recorded as having over-promised. The probe was right; the trainer was passing the teacher a scaler round-trip that is a mathematical no-op and not a numerical one. Fixing it moved the shipped student to 0.0614, above the asserted ceiling.

(c) **"The GPU is unusable for this work."** Reported after `torch.cuda` raised "no kernel image available" on the training host, with the conclusion that the card's compute capability was unsupported. The measurement had been taken with the system interpreter, which carries a CUDA 12.1 build; the project's own virtual environment carries a CUDA 13.0 build that drives the same card correctly. The retraction keeps one real finding: the system interpreter reports `is_available() == True` and then raises at the first kernel launch, so availability is not a capability check.

(d) **A regulatory date "correction" that was itself wrong.** A recorded compliance date was rewritten as stale. Checking the Official Journal text showed the rewrite was premature: a four-month transitional provision gives Article 50(2) systems placed on the market before 2 August 2026 until 2 December 2026. Whether this demonstrator was placed on the market before that date is not recorded in the repository, so which of the two dates binds it is not established.

The common shape is that each retracted claim was *plausible and unverified*, and each was caught by measuring the thing directly rather than by reasoning about it further.

10 Compliance and licensing posture

10.1 Transparency obligations

One component is in scope for AI transparency rules: an agentic panel running a small language model in the visitor's browser. Interaction disclosure is implemented unconditionally rather than behind a toggle, at a measured contrast ratio of 12.42:1. The machine-readable marking obligation is analysed as *not applicable* on a documented chain: the output is free-form text, free-form text cannot carry metadata so watermarking would be the only available layer, and the applicable code of practice excepts text shorter than 200 tokens. The system generates 96 or 160 tokens depending on runtime. Three standing conditions would reopen that analysis and are recorded.

10.2 Data licensing, and a divergence it exposed

Training corpora are not redistributed whole; samples drawn from them are, in some cases a whole test split, with per-corpus licences recorded. One instructive failure: two documents stated that a model was trained from random initialisation *because* pretrained weights carried unresolved redistribution terms, while the artefact published to the model hub recorded exactly those pretrained weights. The licence-safe path had been a non-default flag. It is now the default, and the publication script refuses any model whose manifest records a pretrained source — verified to fire on one and pass on the other. A licence commitment that only a human remembers to honour is not a control.

11 Reproducibility

- **Artefacts available.** 11 models are published with pinned revisions; the demonstrator fetches an immutable commit rather than a moving branch.
- **Held-out samples travel with weights.** Each model ships an evaluation sample in its own repository at the same pinned commit, so the sample and the weights cannot drift apart, together with the expected metric measured on exactly those rows. The browser re-checks its own result against that figure and reports a mismatch instead of rendering it. The one exception is the satellite student, whose event-wise metric cannot be re-scored from independent rows. Every sample is drawn from its model's test partition or test window, and the run log names which; three were drawn from the whole corpus until Round 71.
- **Preprocessing travels with the data.** Every sample carries an explicit preprocessing list applied verbatim by the consumer. This exists because an early evaluation scored a model at 0.4615 against a recorded 0.8782 by omitting one normalisation step — below the majority-class baseline, with no error anywhere.
- **This document.** All tables and figures are generated from the repository's artefacts at build time, so none was transcribed by hand. The numbers quoted in the prose are typed from the records and are not generated.

12 Limitations and further work

- **The satellite metric cannot be made benchmark-comparable** without changing the temporal split, which would be fitting the evaluation to the answer. Three events is not a sample size a bootstrap can rescue.
- **Quantum advantage is not demonstrated** and is not claimed; the measured quantum classifiers lose to the classical baseline on the same data.

- **Structural graph features remain fitted on the training window** rather than computed causally, because recomputing sparse reachability at every timestep is not tractable here. The per-account aggregates are causal; the distinction is stated rather than smoothed over.
- **Fitting the model to the objective it is scored on.** Net benefit currently selects an operating point for an already-fitted model. A sparse integer program can optimise the integrated net-benefit curve *directly*, and the resulting integer-coefficient scoring system is a single encrypted dot product by construction — which fits this architecture better than any of the alternatives evaluated.
- **The verification checklist is 4,783 lines, and it is not published.** It is indexed and every one of its 673 rows has a unique identifier, but its size is itself a finding about how much of this work is institutional memory rather than automation. It is maintained outside the repository because it accumulated the deploy procedure and the internal failure history alongside the checks. The observable outcomes it gates are published in the round register, `docs/deliverable-verification.md`, which has carried the deploy walks since Round 56, and their lessons in `docs/verification-log.md`. An earlier revision of this line said 1,800 lines, which was stale by six hundred; the count is now generated, like the tables' numbers.

13 Conclusion

The engineering claim of this project is narrow and, we think, defensible: it is possible to run genuine post-quantum key establishment, genuine homomorphic linear algebra and genuine garbled-circuit non-linearities in a browser tab, and to report honestly what that costs in model quality.

The methodological claim is broader. 21 classes of silent defect were found in a system whose tests were green, whose types checked and whose aggregate metrics looked reasonable. Every one was found by measuring a specific quantity that nothing had previously measured — the number of distinct classes a model predicts, the agreement between harvested labels and ground truth under a permutation null, the identity of the table a baseline was cited from. None required new tooling. What they required was treating a plausible number as unverified until something checked it, which is a discipline rather than an infrastructure.

14 References

[1] NIST. *FIPS 203: Module-Lattice-Based Key-Encapsulation Mechanism Standard*. August 2024.

[2] NIST. *FIPS 204: Module-Lattice-Based Digital Signature Standard*. August 2024.

[3] D. Moody, R. Perlner, A. Regenscheid, A. Robinson and D. Cooper (NIST). *IR 8547: Transition to Post-Quantum Cryptography Standards*. October 2026, doi:10.6028/NIST.IR.8547; follows the November 2024 initial public draft. Quantum-vulnerable public-key algorithms deprecated after 2030 and disallowed after 2035 (Tables 2 and 4).

[4] NSA. *CNSA 2.0 and Quantum Computing FAQ*, Ver. 2.2, October 2026; and CNSS, *CNSSP 15*, December 2024 — CNSA 2.0 algorithms required in new national security system acquisitions from 1 January 2027 and in all NSS use by 31 December 2031.

[5] Regulation (EU) 2024/1689 (AI Act), Article 50; and Regulation (EU) 2026/1744 (Digital Omnibus on AI), in force 27 July 2026. Article 50(1) applies from 2 August 2026; Article 50(2) systems placed on the market before that date have a transitional period to 2 December 2026.

[6] B. Van Calster *et al.* (STRATOS TG6). "Evaluation of performance measures in predictive artificial intelligence models to support medical decisions." *The Lancet Digital Health* 7(12):100916, December 2025. DOI 10.1016/j.landig.2025.100916.

[7] Z. C. Lipton, C. Elkan, B. Naryanaswamy. "Optimal Thresholding of Classifiers to Maximize F1 Measure." *ECML-PKDD 2014*, LNCS 8725. DOI 10.1007/978-3-662-44851-9_15. Proves that an uninformative classifier's F1-optimal behaviour is to classify every example as positive.

[8] W. Chi, Ş. İ. Birbil. "Learning an Interpretable Risk Scoring System for Maximizing Decision Net Benefit." arXiv:2604.04241, April 2026.

[9] J. Vickers, E. Elkin. Decision curve analysis; net benefit $\mathrm{NB}(p) = \mathrm{TP}/N - (\mathrm{FP}/N) \cdot p/(1-p)$.

[10] S. Kim, K. Choi, H.-S. Choi, B. Lee and S. Yoon. "Towards a Rigorous Evaluation of Time-series Anomaly Detection." AAAI 2022, arXiv:2109.05257 — under point adjustment "even a random anomaly score can easily turn into a state-of-the-art TAD method". This entry cited *TSB-AD* for "a uniform random score ranks 26th of 32 detectors" until Round 33; the rank could not be verified (TSB-AD describes 40 algorithms and recommends VUS-PR), and the entry also attributed TSB-AD to the wrong authors — it is Liu and Paparrizos, NeurIPS 2024 Datasets & Benchmarks, and it carries no arXiv identifier.

[11] K. Kotowski *et al.* "European Space Agency Benchmark for Anomaly Detection in Satellite Telemetry." arXiv:2406.17826. Table 2 (all events) and Supplementary Table 9 (anomalies alone) are different comparators; this project scores anomalies alone.

[12] J. Yang *et al.* "MedMNIST v2." *Scientific Data* 10:41, 2023 — BreastMNIST ResNet-18 AUC 0.901, ResNet-50 AUC 0.857.

[13] E. Altman *et al.* "Realistic Synthetic Financial Transactions for AML Models." NeurIPS 2023 Datasets & Benchmarks (AMLworld).

[14] A. Al Badawi *et al.* "SoK: Private LLM Inference using Approximate Homomorphic Encryption." IACR ePrint 2026/935 — roughly four orders of magnitude slower than plaintext; about 20% of surveyed systems are model-preserving.

[15] Saito & Rehmsmeier. "The precision-recall plot is more informative than the ROC plot." *PLOS ONE*, 2015.

[16] ISO/IEC 28033 (homomorphic encryption). The CKKS part is **Part 3**, arithmetic on approximate numbers, at FDIS stage 50.00 as of 11 June 2026 — one ballot from publication and **not yet published**, so CKKS is not described here as ISO-standardised. Stages checked 10 August 2026; this claim has a shelf life of months.

A Findings and recommendations: where they are kept

The repository's README is a summary since Round 80. The findings it had accumulated moved unchanged into Markdown documents beside the code, where they can be read, searched and compared from one revision to the next; this appendix says what each holds and which of them bear most on what this report claims.

- `docs/findings.md` — the challenges of the QSMPC and QHE workstreams, and every concern and recommendation recorded so far: the published models the pipeline did not run before Round 21, the security score retired in Round 22, the satellite forecaster's retracted result and the leak behind it, the browser model tier, the boundaries of the public demo, and the positions of NSA and BSI on QKD.
- `docs/models-and-benchmarks.md` — the datasets and their licences, the MPC-FHE against QSMPC-FHE benchmark, the model tiers, and what each page shows, including the satellite and fibre key blocks.
- `docs/use-cases.md` — what is and is not new, the positioning (quantum-safe, not quantum-advantage), the four workstreams and the industry use cases.
- `docs/development.md` — the development environment, its GPU, FHE, MPC and Rust/WASM tiers, the dependency advisories and the upstream submodules.
- `docs/deliverable-verification.md` and `docs/verification-log.md` — the record of each round: what was checked, what was found, and what it taught.

Three findings change how the results above should be read. The 1-decoy bound reproduces Rusca et al.'s Table I within 7 percent at 26 and 46 dB once the paper's global attenuation, which already includes the detector, is applied once; the "conservative by 2–3x" carried since Round 47

came from two tests that applied a 50 percent detector on top of it, and Round 82 corrected a third transcription error in that bound, τ_0 on $\delta(n_Z)$ in (A16), which the paper does not print and which had overstated the key. The satellite's finite key, at an error-correction efficiency of 1.16, was 0.55 of Liao et al.'s published key on the same pass and 0.56 of Khmelev et al.'s on theirs under the published Hoeffding statistics; with Chernoff-type statistics, Khmelev's own kind, it is 0.75 and 0.69, so the statistics close 45 percent of the first gap and 29 percent of the second, and the rest is not one shared cause. And the public demo's QKD yields a key length, not key bytes: its quantum channel is simulated.

Reproduction. Revision `a1cd874-dirty`. `make report` regenerates the tables (`report/build_data.py`) and the figures (`report/build_figures.py`, which checks them for overlapping text), then builds the document with two `pdflatex` passes and fails on any line set wider than its box.